# COMS-176: Introduction to .NET Programming

**Programming Activity –** *Final Project*

**Scenario**

To wrap up your learning in this course, you may select from a few different final project applications. Each application option is listed below with details and grading information. Once you select and start a project, you must continue/complete with your choice.

## Bug Tracker

*Create a bug tracking application suited for an audience of 4 to 12 year olds.*

**The application needs to track:**

1. The name of the bug
2. Information about the bug

**This application needs to allow the user to:**

1. Save data about the bugs (text file <recommended> or database)
2. Display bug names in a combo box or some similar control
3. Allow the user to add\edit\delete bugs
4. Include at least five bugs when the program is first run

**This application is graded on the following criteria:**

| Item | Point Value |
|---|---|
| Form design is appropriate for a young child and has a playful design (bright colors, graphics) | 13 |
| Form design includes named controls, tab stops, and good user experience features | 13 |
| Application stores information about bugs in text file or database and effectively displays it to screen | 32 |
| Application includes at least two forms | 18 |
| Competency is shown for variables, lists, methods | 28 |
| Inline documentation is provided for each section of code | 13 |
| Artifacts included per student portfolio directions | 13 |
| **TOTAL** | **130** |

# QUIZZER

Create an application that helps students study for quizzes. The quizzes can be based on any topic, but an example topic would be quiz information about great cartoon shows.

**This application needs to include the following features:**

1. Read quiz questions from a text file
2. Randomly display quiz questions to screen
3. Randomly order answers for each question allowing user to select answer
4. Display whether user was correct or incorrect with answer
5. Display correct answer if user selected incorrect answer
6. Tally how many questions the user has answered, how many have been correct, how many have been incorrect
7. Allow user to add\edit\delete quiz questions and answers (recommend four answers per question with only one being correct)
8. Include at least five quiz questions when program is first run

**Note:**

- It is highly recommended that one form is created for add/edit/delete and another is created for taking the quiz. This will simplify your approach.

**This application is graded on the following criteria:**

| Item | Point Value |
|---|---|
| Form design is appropriate student with an academic appeal (change fonts, backgrounds, etc) | 12 |
| Form design includes named controls, tab stops, and good user experience features | 12 |
| Application stores quiz questions and answers in a text file | 23 |
| Questions to display are randomly selected from pool of questions | 5 |
| Answers are ordered in a random order | 9 |
| Application provides feedback to user about correct/incorrect answers with tally | 9 |
| Application allows user to add\edit\delete quiz questions | 9 |
| Competency is shown for variables, lists, methods | 24 |
| Inline documentation is provided for each section of code | 14 |
| Artifacts included per student portfolio directions | 13 |
| **TOTAL** | **130** |
| *Extra Credit (if all works, this program earns extra credit points)* | *10* |

# My Playful Playlist

***advanced application***

Create an application that creates playlists. This application allows the user to select audio files from their computer and create\edit\delete m3u format playlists.

**This application needs to allow the user to:**

1. Create a new playlist in m3u format
2. Edit a playlist in m3u format
3. Delete a playlist (delete the file from the computer, not the music)
4. Allow the user to enter the name of the playlist (for example, Peanuts Music and check that there are no playlists with that name)
5. Add songs to a playlist from the users computer by using the file browser dialog in Visual Studio
6. Play the playlist in the application
7. Include at least three songs (mp3) in your bin\debug folder and at least two playlists when the program is first run

**Extra knowledge needed beyond what was taught in course:**

- Browsing for songs files using file browser in Visual Studio
- Reviewing the m3u file format (look at Wikipedia)
- Playing audio in a Visual Studio C# application (lookup how to add AxWindowsMediaPlayer to your toolbox and then how to play by URL)

**This application is graded on the following criteria:**

| Item | Point Value |
|---|---|
| Form design is fun in design for a music listener (change fonts, backgrounds, etc) | 12 |
| Form design includes media player control to allow for start, pause, next, previous. | 12 |
| Application creates\edits\deletes playlists | 23 |
| Application allows user to select audio on their computer using file browser | 12 |
| Application plays audio based on playlist selected | 23 |
| Competency is shown for variables, lists, methods | 23 |
| Inline documentation is provided for each section of code | 12 |
| Artifacts included per student portfolio directions | 13 |
| **TOTAL** | **130** |
| *Extra Credit (if all works, this program earns extra credit points)* | *15* |